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Name: ________R. No.____ Date Of Exam: ________ Invig. Sign_____ Total printed pages

Total No. of printed pages: _4_

परमाणु ऊर्जा केन्द्रीय विद्यालय, नरौरा

Atomic Energy Central School Narora

Annual Examination-2019-20

Class : XI

Subject : MATHEMATICS

Time : 3 HRS

Marks : 80

General Instructions:

(i) All questions are compulsory

(ii) The question paper comprises of four sections, A, B ,C and D

(iii) Question 1 to 20 in section A are Multiple Choice and fill in blanks

(iv) Question21 to 26 in Section B carry 2 marks each.

(v) Question 27 to 32 in Section C carry 4 marks each

(vi) Question33 to36 in Section D carry 6 marks each

SECTION - A

Choose and write correct option in each of the following (15 x 1 = 15)

1:Number of the triangle can be formed by joining vertices of hexagon

(a) 6 (b) 20 (c) 12 (d) none of these

2:For disjoint sets A and B ,n(A) =3 and n (B) = 5 then n(A $\cap B$) =

(a) 2 (b) 3 (c) 5 (d) 0

3 :If A ={ 2} B = { 3,4,5 } and C = { 5 ,6} then number of element in A X (B – C) are

(a) 2 (b) 3 (c) 6 (d)1

4:Range of the function f(x) = $\frac{x}{x+2}$ is

(a) R (b) R - {2} (c) R - { -2} (d) none of these

5: If tan A = ½ and tan B= 1/3 ,then A + B is

(a) $\frac{\pi}{6}$(b) π (c)0 (d) $\pi/4$

6:Ifx^n– 1 is divisible by x – k , then the least integral value of k is

(a) 1 (b) 2 (c) 3 (d) 4

(9)

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7; Conjugate of $i^3 - 4$ is

(a) $i^3 + 4$ (b) 4 - i (c) - 4 +i (d) - 4 - i

8: If ${}^nC_{12} = {}^nC_8$ then n is

(a) 20 (b) 12 (c) 8 (d) 4

9 ; The equation of straight line parallel to y axis and at a distance of 4 units its right

(a) x= 4 (b) x= -4 (c) y= 4 (d) y = -4

10 ; The area of the circle centred as (1, 2) and passing through (4, 6)

(a) 5π (b) 10π (c) 25π (d) none of these

11; The equation of YZ plane is

(a x= 0 (b) y = 0 (c) z = 0 (d) none of these

12 :Which of the following is a "statement "

(a) where is your school (b) when you go to Delhi (c) what is the time
(d) sky is red

13: The negation of the statement " 7 is greater than 8 "

(a) 7 is equal 8 (b) 7 is not greater than 8 (c) 8 is less than 7 (d) none of these

14:Total number of the term in the expansion of $(x^3 + 3x^2 + 3x + 1)^{10}$

(a) 11 (b) 21 (c) 31 (d) 30

15 :The value of $\sin (45^0 - A) - \cos(45^0 - A)$ is

(a) 2cos A (b)2 sin A(c) 1 (d)none of these

16 : A polygon has 44 diagonal then number of sides ....

17 : Middle term in the expansion of $(x + \frac{1}{x})^{20}$ is ....

18: Value of $3^{1/2} . 3^{1/4} .3^{1/8}$.................infinite term is ....

19: Distance of the point (3,4,5) from origin is......

20: $i^n + i^{n+3} + i^{n+2} + i^{n+1} =$ ....

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SECTION B (2x6 =12)

21: Solve for $x \in C$: $2x^2 + 3ix + 2 = 0$

22: Solve the equation sin x+ cos x =0

23 : Evaluate $\lim_{x\to 0}(\sin 3x + \sin x)/x$

24: Find the domain of the function $f(x) = \sqrt{(9/4 - x^2)}$

25: If $y = \sin^2(\cos\sqrt{x})$,find dy/dx

26 :Find the eccentricity for the ellipse $2x^2 + 5y^2 = 10$

SECTION C (4x6 =24)

27 : Solve the following system of linear inequations graphically

$2x + y \geq 4,\quad x + y \leq 3,\quad 2x - 3y \leq 6,\ x \geq 0, y \geq 0$

28 Prove, by using principle of mathematical induction, for all natural numbers

$X^{2n} - y^{2n}$ is divisible by $(x + y)$

29: Find the image of the point (- 1, 2) with respect to the line mirror $2x - y + 1 = 0$

OR

Find the equation of the circle passing through the points (-3,1),(6,4) and (2,-6)

30: Four cards are drawn from a pack of 52 playing cards. Find the probability of

drawing 3 clubs and 1 heart

OR

If a pair of dice is thrown once find the probability of sum is 8 or 9

31: If all the letters of the word NARORA be arranged as in a dictionary ,what is the rank of word. NARORA

OR

If all the letters of the word AGAIN be arranged as in a dictionary ,what is the fiftieth word

32: Calculate the standard deviation for the following data

C I	0 -10	10 - 20	20 - 30	30- 40	40 - 50	50 - 60	60 -70	70 - 80
f	5	8	7	12	28	20	10	10

SECTION D (6 x4 = 24)

33 : In a town of 10,000 families it was found that 40% families buy news paper A ,

20% families buy newspaper B, 10% families buy newspaper C,5% families buy A and B, 3% families buy B and C, 4% families buy A and C .If 2% families buy all three newspapers .Find the number of families which buy at least two papers

34: In any triangle ABC, prove that:

$a^3 \sin(B - C) + b^3 \sin(C - A) + c^3 \sin(A - B) = 0$

:OR

Prove that : $\frac{(\sec 8A - 1)}{(\sec 4A - 1)} = \tan 8A . \cot 2A$

35 ;C(n, r - 1) : C (n, r) : C (n ,r + 1) = 1 : 7 : 42 , find n and r

OR

If P(n, r) = P (n, r+1) and C(n, r) = C (n, r -1)

,Find the value of n and r

36:Find the sum to n-terms of the series

$1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots\dots\dots\dots$

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ATOMIC ENERGY CENTRAL SCHOOL, NARORA

ANNUAL EXAM 2019-20

CLASS XI

TIME:3 HRS

SUB:BIOLOGY

MARKS:70

Questions 1-5(1 Mark each), 6-12(2 Marks each), 13-24(3 Marks each),25-27(5 Marks each)

1.Pick out the protest which caused the red tide in ocean:

Euglena,Trypanosoma, Gonyaulax, paramecium

2.Deficiency symptoms of which of the following elements appear first in younger parts of the plant?

(a) Nitrogen (b) Potassium (c) Magnesium (d) Calcium

3.Expand ICBN

Or

Pepsinogen is an inactive enzyme secreted by gastric glands of the stomach. How is it activated?

4. Name the type of bond by which the monomers in the following are held together:

(a) Deoxyribonucleic acid (b)Cellulose

Or

Name (a) the smallest and (b)the longest cells in human body.

5.Name the bile pigements.

6. How does aldosterone function in our body?

Or

Mention the role of haemoglobin in the transport of respiratory gases in human beings.

7. What are porins? What role do they play in diffusion?

8. Draw a labeled diagram of cardiac muscle tissue.

9. Write any four points of differences between chondrichthyes and Osteichthyes.

10.(i) What is the nature of ovary in

(a) Hypogynous flowers and

(c) E (d) I

(b) Epigynous flowers?

(ii) Name the types of phyllotaxy shown in (a) Guava (b) Hybiscus

11. What are plasmids? Mention their significance in prokaryotic/bacterial cells.

Or

Fill in the blanks A,B, C and D in the table with regard to the events in the different phases of cell cycle.

Cell cycle stages	Characteristic event
A	Nucleic acids double in quantity
Anaphase	B
Metaphase	C
D	Chromatids or daughter chromosomes start uncoiling

12. Distinguish between heart wood and sap wood.

13. Name the hormone secreted by parafollicular cells of thyroid. Mention the condition when it is secreted and its function.

Or

Draw the diagram of human eye as seen in a V.S and label six parts in it.

14.(a) What is the role of troponin and F-actin during contraction in striated muscle of humans?

(b) Why are human ribs described as bicephalic?

15. (a) Draw the structure of amino acid, Alanine.

(d) Differentiate between Ligases and Lyases.

16.(a) Why are Deuteromycetes called imperfect fungi?

(b) What do the terms phycobiont and mycobiont signify.

(c) Name the phycomycete that is parasite on mustard.

Or

(a)How is the gametophyte of Bryophytes different from that of Gymnosperms?

(b) Give two examples of Liverworts.

17. Explain the transmission of nerve impulse across a chemical synapse.

18. Give a comparision between C_3 and C_4 pathways of photosynthesis.

Or

...tendency?

(d) I

(a)Define a joint.

(b)Name the type of synovial joint between:

(i) Atlas and axis

(ii) Carpel and metacarpal of tongue

(iii) Humerus and pectoral girdle

(iv) Carpels

19. (a) Enumerate the key features of meiosis.

(b) At what phase of cell division do the centrioles duplicate in animal cells?

20.(a) All vertebrates are chordates, but all chordates are not vertebrates. 'Justify your statement.'

(b) How important is the presence of air bladder in pisces?

21. Describe the steps in the formation of root nodules in a legume plant.

22. (a)Where is Sino-atrial node located in human heart? Why is it called the pace maker of our heart?

(b) How is the rate of heart beat determined from the ECG?

23. Explain micturition.

24. Draw the floral diagram of the family commonly called potato family and write its floral formula.

25. Where does non-cyclic photo phophorylation take place? Describe it. Why is it called so?

Or

Where does glycolysis occur in a cell? Describe the sequence of reactions in it. Mention the end products.

26. Who proposed the fluid mosaic model of plasma membrane? Describe the same with the help of a labeled diagram.

Or

What is a centromere? How does the position of centromere form the basis of classification of chromosomes? Support your answer with diagrams showing the position of centromere on different types of chromosomes.

27.(a) Describe how plants like *Sorghum* fix the carbon dioxide into glucose.

(b) How does water stress affect photosynthesis?

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ATOMIC ENERGY CENTRAL SCHOOL, NARORA

ANNUAL EXAM 2019-20

CLASS-XI

M.M-70

SUBJECT-CHEMISTRY

TIME- 3 Hr

General Instructions:

(a) All questions are compulsory.
(b) Section A: Q.no. 1 to 20 are very short answer questions (objective type) and carry 1 mark each.
(c) Section B: Q.no. 21 to 27 are short answer questions and carry 2 marks each.
(d) Section C: Q.no. 28 to 34 are long answer questions and carry 3 marks each.
(e) Section D: Q.no. 35 to 37 are also long answer questions and carry 5 marks each.
(f) There is no overall choice. However an internal choice has been provided in two questions of two marks, two questions of three marks and all the three questions of five marks weight age. You have to attempt only one of the choices in such questions.
(g)Use log tables if necessary, use of calculators is not allowed.

SECTION-A

Read the given passage and answer the questions 1 to 4 that follow:

Magnesium is present in the Earth's crust to about **2.5** percent by mass. The typical ores of magnesium are dolomite ($CaCO_3.MgCO_3$) and epsomite ($MgSO_4.7H_2O$). Sea water is another source of magnesium. Roughly 1 kilogram of sea water contains **1.3** g of magnesium as magnesium chloride. It can be obtained by carrying out the electrolysis of molten $MgCl_2$.

Q.1- What happens when magnesium is heated in air?

Q.2- Which product is obtained when magnesium oxide is dissolved in water?

Q.3-What is use of product obtained by dissolution of magnesium oxide.

Q.4- What is the significance of magnesium in human body?

Question from 5 to 10 are multiple choice question.

Q.5- Hybridisation of chlorine atom in ClO_2^- is:

(a) sp^3 (b) sp^2 (c) sp (d) None of these

Q.6-The volume of gas is reduced to half from its original volume. The specific heat will-

(a) Reduce to half (b) be doubled (c) remain constant (d) increase four times

Q.7-Which of the following elements does not show disproportionation tendency?

(a) Cl (b) Br (c) F (d) I

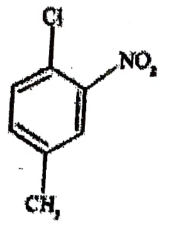
Q.8-The low density of ice compared to water is due to.

(a) Hydrogen bonding interactions (b) dipole – dipole interactions
(c) dipole-induced dipole interactions (d) induced dipole – induced dipole interactions

Q.9- Quartz is a crystalline variety of

(a) Si (b) SiO_2 (c) Na_2SiO_3 (d) SiC

Q.10- The IUPAC name for



(a) 1-Chloro-2-nitro-4-methylbenzene
(b) 1-Chloro-4-methyl-2-nitrobenzene
(c) 2-Chloro-1-nitro-5-methylbenzene
(d) *m*-Nitro-*p*-chlorotoluene

Questions 11 to 15 are one word answers:

Q.11- What is the general outer electronic configuration of f – block elements?

Q.12- The mass number of an element is twice its atomic number. If there are four electrons In 2p orbitals, write the electronic configuration of the element and name it.

Q.13- Compressibility factor, of a gas is given as $Z=PV/nRT$

(i) What is the value of Z for an ideal gas?

(ii) For real gas, what will be the effect on value of Z above boyle's temperature?

Q.14- Define the Entropy.

Q.15- What is BOD?

Questions 16 to 20: In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as:

(A)Both assertion and reason are correct statements, and reason is the correct explanation of the assertion.

(B) Both assertion and reason are correct statements, but reason is not the correct explanation of the assertion.

(C) Assertion is correct, but reason is wrong statement.

(D) Assertion is wrong, but reason is correct statement.

Q.16- Assertion: Boron has a smaller first ionisation enthalpy than beryllium.

Reason: The penetration of a 2s electron to the nucleus is more than the 2p electron Hence 2p electron is more shielded by the inner core of electrons than the 2s electrons.

Q.17- Assertion: At constant temperature, PV vs V plot for real gases is not a straight line.

Reason: At high pressure, all gases have $Z > 1$ but at intermediate pressure most gases have $Z < 1$.

Q.18- Assertion: Combustion of all organic compounds is an exothermic reaction.

Reason: The enthalpies of all elements in their standard state are zero.

Q.19- Assertion: In the reaction between potassium permanganate and potassium iodide, permanganate ions act as oxidising agent.

Reason: Oxidation state of manganese changes from +2 to +7 during the reaction.

Q.20- Assertion: Energy of resonance hybrid is equal to the average of energies of all canonical forms.

Reason: Resonance hybrid cannot be presented by a single structure.

SECTION-B

Q.21-Calculate the number of atoms in:

a) 0.25 mole atom of carbon b) 0.20 mole molecules of oxygen.

Q.22- A microscope using suitable photons is employed to locate an electron in an atom with in a distance of 0.1×10^{-10} m. What is the uncertainty involved in the measurement of

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its velocity?

Q.23- Explain the screening effect.

Q.24- (i) Why does the boiling point of liquid rise on increasing pressure?

(ii) Why does viscosity of liquids decrease as the temperature is raised?

Q.25- Explain the Lewis acid base concept.

OR

Answer the following:

(i) Acetic acid is highly soluble in water but still a weak electrolyte. Why?

(ii) The value of the equilibrium constant is less than zero. What does it indicate?

Q.26- Balance the following equations by ion electron method:

$MnO_4^-{}_{(aq)} + I^-{}_{(aq)} \rightarrow MnO_2(s) + I_{2(S)}$ (in basic medium)

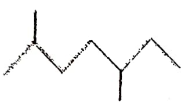
OR

(A) Calculate the oxidation number of following underlined species.

(i) K<u>Mn</u>O_4 (ii) <u>Cr_2</u>O_7^{-2}

(B) What is the decomposition reaction? Give an example.

Q.27- Write the IUPAC name of following:

(a)  (b) $Cl_2CH\ CH_2OH$

SECTION-C

Q.28- A crystalline compound when heated anhydrous by losing 51.2% of the mass. On analysis, the anhydrous compound gave the following percentage composition: Mg=20%, S=26.66% and O=53.33%. Calculate the molecular formula of the anhydrous compound. The molecular mass of anhydrous compound is 120 u.

Q.29- A neutral atom has 2K electrons, 8L electrons and 6M electrons. Predict from this:

(i) Its atomic number (ii) Total number of s-electrons (iii) Total number of p-electrons

Q.30-Using molecular orbital theory compares the relative stabilities and magnetic properties of O_2^+, O_2^- and O_2^{2-} species.

OR

a) Predict the shapes of the following molecules:

(i) ClF_3 (ii) IF_7

b) Explain the following terms with suitable examples.

i) Dipole moment (ii) Sigma and Pi bonds

Q.31- What is the Hess's law. Write also about its applications.

Q.32- (a) What is the hard and soft water? What are the causes of hardness of water?

(b) What is the occlusion of hydrogen?

Q.33- Answer the following:

(i) What is the inert pair effect? (ii) What are silicones?

(iii) Discuss the structure of diborane.

OR

Why is CO_2 a gas at room temperature and SiO_2 is a solid?

Q.33- Explain the following:

1) What do you understand by acid rain? Also write the reactions.

2) What are Greenhouse Gases? Name them.

...the increase in temperature viscosity of ... (b)Liquid decreases and of gases increases

...gases decreases (d)Both liquid and gases decreases

SECTION-D

Q.35- For the exothermic formation of sulphur trioxide from sulphur dioxide and oxygen in the gas phase: $2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$ $K_p = 40.5\ atm^{-1}$ at 900 K and $\Delta H = -198kJ$
Answer the following

1) At room temperature will K_p be great than, less than or equal to K_p at 900 K.
2) How will the equilibrium be affected if the volume of the vessel containing the three gases is reduced, keeping the temperature constant? What happens?
3) What is the effect of adding 1 mole of He(g) to a flask containing SO_2, O_2 and SO_3 at equilibrium at constant temperature?

(i) At 700K, the equilibrium constant for the reaction $H_{2(g)} + I_2 \rightleftharpoons 2HI_{(g)}$ is 54.8. If 0.5 mol L^{-1} of HI(g) is present at equilibrium at 700 K, what are the concentration of $H_{2(g)}$ and $I_{2(g)}$? Assume that we initially started with $HI_{(g)}$ and allowed it to reach equilibrium at 700 K.

OR

1) Hydrogen gas is obtained from the natural gas by partial oxidation with steam as per following endothermic reaction: $CH_{4(g)} + H_2O \rightleftharpoons CO_{(g)} + 3H_{2(g)}$
Write expression for K_p for the above reaction. How will value of K_p and composition of equilibrium mixture be affected by:
a) Increasing the Pressure (b) Increasing the temperature (c) using a catalyst?

2) Explain reaction Quotient in detail and how it is used in predicting the direction of reaction.

Q.36- (a) Explain the following and give the 2 example of each
i) Position isomerism (ii) functional isomerism

(b) Answer the following
(i) Electrophilic substitution reaction (S_E reaction)
(ii) Test for unsaturation
(iii) β-elimination reaction

OR

A. Explain the following with suitable examples.
1) Electrophiles 2) Nucleophiles 3) Inductive effect

B. What is the resonance effect? Write also about its type with suitable examples.

Q.37- Answer the following:
1) Halogenation of benzene 2) Nitration of benzene 3) Sulphonation of benzene
4) friedel crafts alkylation 5) Huckel rule

OR

1) Convert the following:
a) Ethyne to ethane b) Methane to tetra chloromethane c) Hexane to benzene

2) Answer the following
a) Wurtz reaction b) Markovnikov rule.

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Name: ____________________ R. No.____ Class/ Sec:____ Date:__________ Invig. Sign______

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ATOMIC ENERGY CENTRAL SCHOOL, NARORA

CLASS – XI

Annual Examination (SESSION- 2019-20)

SUBJECT – PHYSICS

TIME ALLOWED – 3 hours

MAX. MARKS - 70

General Instructions:

1. All questions are compulsory.
2. There is no overall choice. However, internal choice has been provided in one question of 2 marks, two question of 3 marks & in each question of 5 marks.
3. Questions 1 to 20 carry 1 mark each.
4. Questions 21 to 27 carry 2 marks each.
5. Questions 28 to 34 carry 3 marks each.
6. Questions 35 to 37 carry 5 marks each.
7. Draw neat labeled diagram whenever necessary.

SECTION A (OBJECTIVE TYPE/VSA)

1. 1 AU is the distance between
   (a) Sun and Earth (b) Earth and Moon (c) Moon and Sun (d) Mercury and Sun
2. The slope of tangent drawn on v-t graph at any instant of time is equal to the
   (a) Instantaneous Acceleration (b) Velocity (c) Impulse (d) Momentum
3. Free fall of an object in vacuum is case of motion with
   (a) Uniform velocity (b) Uniform acceleration (c) Variable acceleration (d) Uniform speed
4. A cyclist bends while taking turn to
   (a) Reduce friction (b) generate required centripetal force
   (c) reduce apparent weight (d) Reduce speed
5. A body is being raised to a height h from surface of earth. What is the sign of work done by applied force and gravitational force respectively
   (a) Positive, Positive (b) Positive , Negative (c) Negative, Positive (d) Negative, Negative
6. Moment of a couple is called
   (a) Angular momentum (b) Force (c) Torque (d) Impulse
7. Kepler's second law is a consequences of
   (a) Conservation of Energy (b) Conservation of linear momentum
   (c) Conservation of angular momentum (d) Conservation of mass
8. With the increase in temperature viscosity of
   (a) Liquid increases and of gases decreases (b)Liquid decreases and of gases increases
   (c) Both liquid and gases increases (d)Both liquid and gases decreases

9. Zeroth law of thermodynamics leads to the concept of

(a) Internal energy (b) Heat content (c) Pressure (d) Temperature

10. The total energy of a simple harmonic oscillator is directly proportional to

(a) Square of Amplitude (b) Amplitude (c) velocity (d) frequency

11. Show graphically how density of a substance varies with temperature.
12. Which is more elastic –steel or rubber? why?

OR

Which is more elastic --water or air? why?
13. Write down relationship between torque and angular momentum
14. Angle of repose for a rough inclined plane is 60°. Find the coefficient of friction for this plane.
15. Write down the dimensional formula for (i) Gravitational constant (ii) Linear Mass Density
16. What is the velocity of a projectile projected with speed u at an angle θ to the horizontal at the highest point of its flight?
17. Draw L-T graph for simple pendulum .Where the symbols have their with usual meanings.
18. If earth be at one half of its present distance from sun .How many days will there be in a year

OR

Value of g is largest at the poles of the earth. Why?
19. How are the orbital and escape velocity of a satellite related to each other.
20. A square frame of wire with side L is dipped in liquid. On taking out a membrane is formed. If surface tension of liquid is S, then find out force acting on the frame?

SECTION - B (2 marks)

21. A stone thrown vertically upward with velocity 'v', comes down after time 't'. Draw the graph showing the variation of speed v/s time and velocity vs time for its entire motion
22. Write laws of limiting friction.

OR

Show that angle of friction is equal to angle of repose.
23. Derive the relation between α and γ where α and γ are coefficient of linear and volume thermal expansion .
24. Write the formula for the energy of one mole of a gas. The energy of one mole of a gas is 264 J at 27°C. At what temperature kinetic energy will become ¾ of its value at 27 °C?
25. To a person standing on the plat form the frequency of the whistle of a train heard by him is respectively 2.4 kHz and 1.6 kHz when the train approaches and recedes away from platform. If velocity of the sound wave is 300 m/s, find the velocity of the train
26. State and prove work energy theorem for constant forces.

OR

Show that mechanical energy of a free falling body is constant
27. What are the amplitude, angular frequency, frequency and time period of a body executing SHM represented by equation $x(t) = 300 \sin 6\pi t$. Where x in metre and t in seconds

SECTION -C (3 marks)

28. A stone is thrown from the top of a tower at an angle 30° down with the horizontal with a velocity of 10 ms^{-1}. If the stone strikes the ground at a distance of 17.3 m from the base of the tower. Calculate the height of the tower. ($g = 10\ ms^{-1}$)
29. Show that Newton's second law of motion is the real law of motion.

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OR

State and prove law of conservation of linear momentum. How is it remain conserved when a bullet is fired from a gun. Explain

30. Define elastic collision and show that relative velocity of approach is same as that of separation in elastic collision in one dimension.
31. Determine the height of geostationary satellite .
32. What is first law of thermodynamics. Give its mathematical form in case of isothermal, adiabatic, isobaric and isochoric process.
33. Derive an expression of potential energy per unit volume for a stretched wire.
34. State parallel axis theorem on moment of inertia and Find the moment of inertia of a uniform rod of length 'L' and mass 'm' about an axis perpendicular to it and passing through a point which is at a distance 3L/4 from its one end.

OR

A circular ring and a circular disc of same mass and same radius are rotating about their diameters with same angular speeds .Which of them will require more torque to be applied to stop them. Give reason in support of your answer.

SECTION - D (5 Marks)

35. (a)State and prove Bernoulli's theorem.
   (b) How does limitations of Bernoulli's theorem?

   OR

   (a) Define capillarity. Give reason for the rise of liquids in capillary Tube.
   (b) Derive ascent formula and discuss the case of insufficient length of capillary tube.
36. What is the need of Banking of road. Derive it expression for velocity of car at bank road. Write its optimum condition for velocity of car.

   OR

   Define centripetal force.
   Establish the relation for the maximum velocity of a car of mass 'm' turning on a level road turn of radius 'R'.
   A car speeding at 72 km/h on a level road takes a sharp circular turn of radius 10 m without reducing the speed. The coefficient of static friction between the road and tyre is 0.5. Will the car slip while taking the turn? Explain briefly.
37. Define beats and beat frequency.
   Show that when two sound sources are vibrating together the beatfrequency is equal to the difference of the frequencies of the sources.

   OR

   Define stationary waves. Derive the equation for the stationary wave in a stretched wire and discuss the modes of vibration in it with suitable diagrams.

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ATOMIC ENERGY CENTRAL SCHOOL, NARORA
ANNUAL EXAMINATION 2019-2020
SUBJECT-ENGLISH
CLASS- XI

NAME..................................CLASS........ SUB..............DATE........... PAGE NO............

TIME: 3hrs M.M. 80

General Instructions:

1. This paper has three sections: A, Band C. A. All the sections are compulsory.
2. Read the instructions carefully and follow them.
3. Do not exceed the prescribed word limit.

SECTION- A: READING (20 Marks)

1. Read the passage carefully and answer the questions that follow: 12

1. The New York is a time for resolutions. Mentally, atleast, most of us could compile formidable lists of do's and don'ts. The same old favorites recur year in year out with monotonous regularity. We resolve to get up earlier each morning, eat less, find more time to play with the children, do a thousand and one jobs about the house, be nice to people we don't like, drive carefully and take the dog for a walk every day. Past experience has taught us that certain accomplishments are beyond attainment. If we remain deep-rooted liars, it is only because we have so often experienced the frustration that results from failure.
2. Most of us fail in our efforts at self-improvement because our schemes are ambitious and we never have time to carry them out. We also make the fundamental error of announcing our resolution to everybody so that we look even more foolish when we slip back into our bad old ways. Aware of these pitfalls, this year I attempted to keep my resolutions to myself. I limited myself to two modest ambitions: to do physical exercise every morning and to read more every evening. An all-night party on New Year's Eve provided me with a good excuse for not carrying out either of these new resolution on the first day of the year, but on the second, applied myself assiduously to the task.
3. The daily exercise lasted only eleven minutes and I proposed to do them early in the morning before anyone had got up. The self discipline required to drag myself out of bed eleven minutes earlier than usual was considerable. Nevertheless, I managed to creep down into the living-room for two days before anyone found me out. After jumping about on the carpet and twisting the human frame into uncomfortable positions, I sat down at the breakfast table in exhausted condition. It was this that betrayed me. The next morning the whole family trooped in to watch the performance. That was really upsetting but I fended off the taunts and jibes of the family good humouredly and soon everybody got used to the idea. However, my enthusiasm waned, the time I spent exercises gradually

diminished. Little by little the eleven minutes fell to zero. By January 10^{th} I was back to where I had started from. I argued that if I spent less time exhausting myself at exercises in the morning I would keep my mind fresh for reading when got home from work. Resisting the hypnotizing effect to television, I sat in my room for a few evenings with my eyes glued to a book. One night, however, feeling cold and lonely, I went downstairs and sat in front of the television pretending to read. That proved to be my undoing, for I soon got back to the old habit of dozing off in front of the screen. I still haven't given up my resolution to do more reading. In fact, I have just bought a book entitled 'How to read a Thousand Words a Minute'. Perhaps it will solve my problem, but I just haven't had time to read it.

1.1. Answer the following questions by choosing the most appropriate option: (1x6=6)

1. According to the writer, past experience of resolutions has taught us:
   a. frustration results from failure
   b. certain accomplishments are beyond attainment
   c. new year is a time for resolutions
   d. failure are a part of life
2. Most of us fail in our efforts at self-improvement because.....
   a. our schemes are too ambitious
   b. we never have time to carry them out
   c. we announce our resolution to everybody
   d. all of these
3. It is a basic mistake to announce our resolution because.....
   a. we have no excuse to revert back to our earlier life
   b. we can't be nice to people we don't like
   c. we look more foolish when we slip back to our old ways
   d. none of these
4. The writer did not carry out his resolutions on New Year's Day because.........
   a. he had attended an all night party
   b. he was glued to the TV
   c. the exercise was only for eleven minutes
   d. everyone was awake and watching him
5. The writer thought of keeping his mind fresh by:
   a. watching TV
   b. exercising for eleven minutes
   c. reading
   d. finding more time to play with children
6. The word which is similar in meaning to 'decrease' is........
   a. formidable
   b. monotonous

c. attainment

d. diminish

1.2. Answer the following questions: (1x6=6)

1. What do people normally resolve to do every New Year?
2. Why do most of us fail in our efforts at self improvement?
3. Why could the writer not carry out his resolution on the very first day of New Year?
4. Why is announcing our resolution a fundamental error?
5. 'The next morning the whole family trooped in to watch the performance'. What performance is the writer talking about?
6. Which book did the writer buy?

2. Read the given passage and answer the questions that follow: 8

1. Good health is the soundness of body, mind and soul. It is that condition in which the body and mind duly discharge their functions. Good health helps everyone to be creative and work for the welfare of the society. He has to be, therefore free from diseases, or, if he suffers from a disease, he needs treatment not only for that particular disease but for the whole body, mind and soul. In other words, a doctor must have a sound approach to the patient.
2. In the days of specialization, doctors cure a patient only for his immediate disease, treating it in isolation. Such an approach goes against the old truism that "prevention is better than cure."Dr. Hedge, insight into the overall causes of simple and serious disease and describe the methods of preventing them without having to take medicine but by exercising, self-control on food consumption, smoking and intake of alcoholic drinks. He also advocates regular yoga exercise and meditation to keep fit but at the same time warns against over- indulgence.
3. Hedge stresses the importance of laughter because laughter induces secretion of good catecholamines which are hormones produced by the adrenal gland; they can be both beneficial and harmful. They are harmful while a person is angry but beneficial if he laughs. For healthy living, "laughter proves to be a great boon", Hegde says.
4. In these days of hectic life, full of stress, strain and emotions, everyone needs to lead a relaxed life. The best form of relaxation, according to Hegde, is sound sleep. "Restful sleep at night will recharge our batteries for the following day's fruitful endeavours." he says. There is no use lying on the bed and tossing without sleep. Under those circumstances it is better to read or engage oneself in some light mental activity which is useful but never seek refuge in alcoholic drinks or pills for inducing sleep as these are not refreshing for an individual. For normal human being, eight hours of sleep a day is sufficient to keep him free from stress. More than nine to ten hours of sleep a day would produce increased muscle protein loss in the body and might even shorten one's life.
5. Society must be sympathetic to drugs addicts, understand their problems and give help to them by motivating and keep them busy. Alcoholic anonymous may help addicts kick

their habit of drug abuse and excessive drinking. Hedge claims that cigarette smoking causes more premature deaths than all the other "killers" put together AIDS, cocaine, heroin, alcohol, fire, automobile accidents, homicide and suicide. Ninety percent of cancer victims are heavy smokers.

6. The author stresses on the right type of diet and daily exercise for a healthy living. He has also talked of the dangers of over-exercising in these days of fitness mania.

2.1 On the basis of your reading of the above passage make notes on it using recognizable abbreviations wherever necessary. Give a suitable title to the passage. 5

2.2 Write a summary of the above passage based on your notes. 3

SECTION B: WRITING & GRAMMAR (30 Marks)

3. As the principal of your school, write a notice in 50 words informing the students about the special coaching in Cricket, Basketball and Tennis during the summer vacation. 4

Or

Gopal Sharma/Garima Sharma have one flat for sale. Draft an advertisement for him to be inserted in the classified columns of the local newspaper. Invent necessary details.

4. Roadside vendors occupy most of the space on roads disturbing the flow of traffic and causing difficulties even to pedestrians. Write a letter to the Editor of a National Daily expressing your views and suggesting ways and means to curb this problem. You are Rohan/Rachna, 45 Navketan Apartments, Lucknow. 6

Or

You are Mr. Manohar, Principal of New Horizon Senior Secondary School, Chennai. Write a letter to the manager of Samsung Electronic Appliances, Pvt. Ltd., Hyderabad placing an order for headsets with Mikes and MP3 players for the language lab of your school. Give necessary specifications- date of delivery, terms and conditions etc.

5. You are Aditya/ Aradhya living at 22, Sector 12, RK Puram, New Delhi. You have seen an advertisement in the newspaper for the post of a 'Software Engineer' in National Software Company. Write an application with complete bio-data to the Manager of the Company. 6

Or

The Canteen of your school is supplying oily and sweet food. Write a letter to the Principal of your school requesting him/her to improve the quality of the food sold by the canteen.

6. Rajiv Malhotra was much inspired by watching the celebration of India's 50 years of independence at the India Gate and Rajpath. Imagine yourself as Rajiv, write an article on 'India of My Dream' for your school magazine in about 200 words. 8

Or

You are disturbed by the fact that the students of today undergo a lot of stress and pressure in life. You have found peace and relaxation by practicing Yoga. You wish to share your experience with the students of your school. Prepare a speech to be given in the morning assembly on "The Benefits of Yoga". You are Suman/ Sumit, the school Captain.

7. The following passage has not been edited. There is an error in each line. Write the incorrect word and the correction in your answersheet. 1/2x6=3

	Incorrect	Correct
Kut –kut was an hardworking	....	....
squirrel. She lived on a tree hole.	....	....
She was over the impression that	....	....
she was enough food in store for the	....	....
bad days until she find that	....	....
someone had been stealing his nuts.	....	....

8. Below is a set of instructions for washing clothes. Imagine that you have completed this procedure. Complete the following paragraph reporting what you did: 1/2x4=2
   a. Pour water in the washing machine upto the marked level.
   b. Add sufficient detergent powder to give foam.
   c. Put in the fine clothes first and switch on the machine.
   d. Switch off the machine after five minutes.

   Water was poured in the washing machine upto the marked level, sufficient powder (a)........... to give rich foam. The fine clothes (b)...........and the machine (c)..........After five minutes the washing machine (d)..............

9. Re- arrange the following words/phrases into meaningful sentences: 1/2x2=1
   1. number one/is/eradicate/to/poverty/millennium development/goal
   2. my/winning/a/medal/I/told/him/until/about/not/known/he/had

SECTION-C: TEXT BOOKS & LONG READING TEXT (30 marks)

10. Read the following extract and answer the questions that follow:(any two parts) 1x2=2

A sweet face,
My mother's that was before I was born
And the sea, which appears to have changed less
Washed their terribly transient feet"

a. How was the mother before the poet's birth?
b. What /who has not changed much?
c. What does the last line imply?

Or

I descend to lave the drought, atomies, dust-layers of
the globe.
And all that in them without me were seeds only, latent,
unborn;
And forever, by day and night, I give back life to my
Own origin,
And make pure and beautify it;

a. Who is 'I'? What does I do on descending?
b. How does 'I' affect those that have seeds in them?

Ⅎ/~

c. Explain the meaning of the last line.

11. Answer any of the five of the following questions: 2x5=10

   a. How did the sparrows express their sorrow when the author's grandmother died?
   b. What problem did Carter face when he reached the mummy? How he find a way out?
   c. How did the old lady satisfy herself about Ranga?
   d. Where did Millie send Taplow? What was her purpose?
   e. Why did Albert hate his lodgings? What would he do to divert his mind?
   f. Write two instances of irony from the poem 'The Tale of Melon City'.
   g. What does the notice "The world's most dangerous animal" at a cage in the zoo at Lusaka, Zambia, signify?

12. "We have not inherited this earth from our forefathers we have borrowed it from our children." Elucidate. 6

Or

What message does, "We're Not Afraid to Die....if We Can All Be Together" suggest?

13. What is the main idea of the play 'Mother's Day'? Has it been brought out effectively by the writer? 6

Or

'I have done something, Oh, God! I've done something real at last. Why does Andrew say this? What does it mean?

14. Draw a character sketch of Khushwant Singh's grandmother as portrayed by him in the lesson 'The Portrait of a Lady'. 6

Or

Give a character sketch of Crocker Harris from what you have read from 'The Browning Version.'

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